Test of Hypothesis (1)

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Hypothesis

- A **hypothesis** is an assumption/statement to be tested.



Statistical hypothesis

- A statistical hypothesis is some statement about a population parameter or about the probability distribution characterizing a population which we want to verify on the basis of sample information.

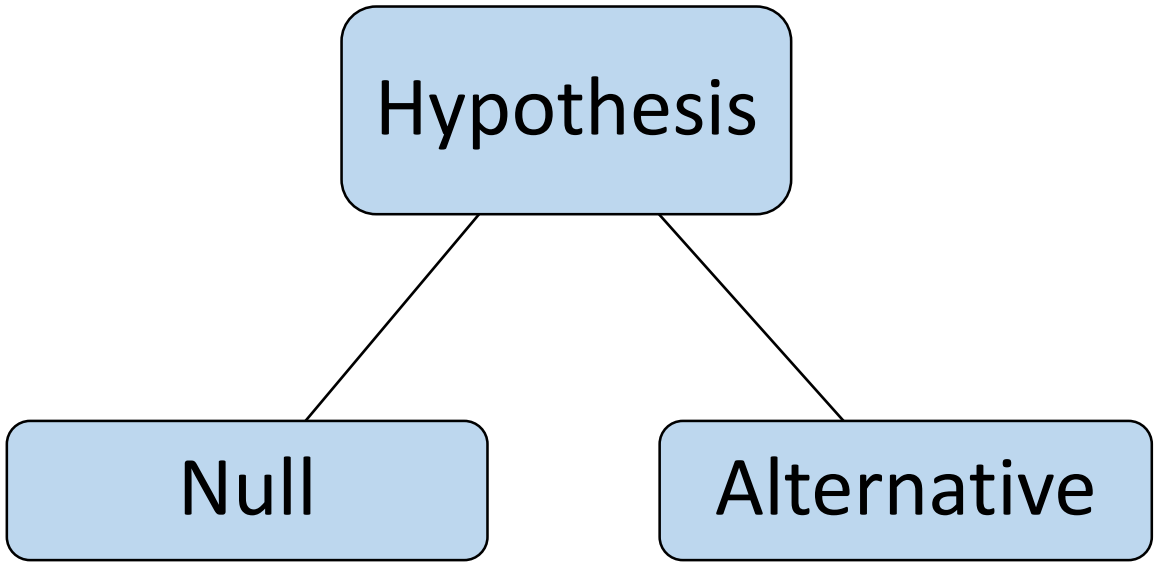
- Example: The average customer satisfaction rating is 4 out of 5.

Getting population data is hard because there's a lot of it.

Getting sample data is easy

Statistical hypothesis

- Two types of statistical hypothesis



Null hypothesis

- A null hypothesis is a statistical hypothesis that contains a statement of equality

- $\leq or = or \geq$

- Denoted by H_0

Alternative hypothesis

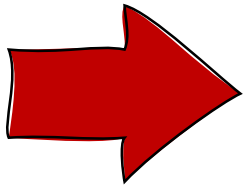
- An alternative hypothesis is a statistical hypothesis that contains a statement of inequality

- $< or \neq or >$

- Denoted by H_1

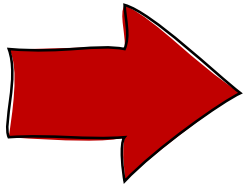
Hypothesis

The new drug has no effect on reducing blood pressure compared to the placebo



H_0

The new drug reduces blood pressure more effectively than the placebo



H_1

Example

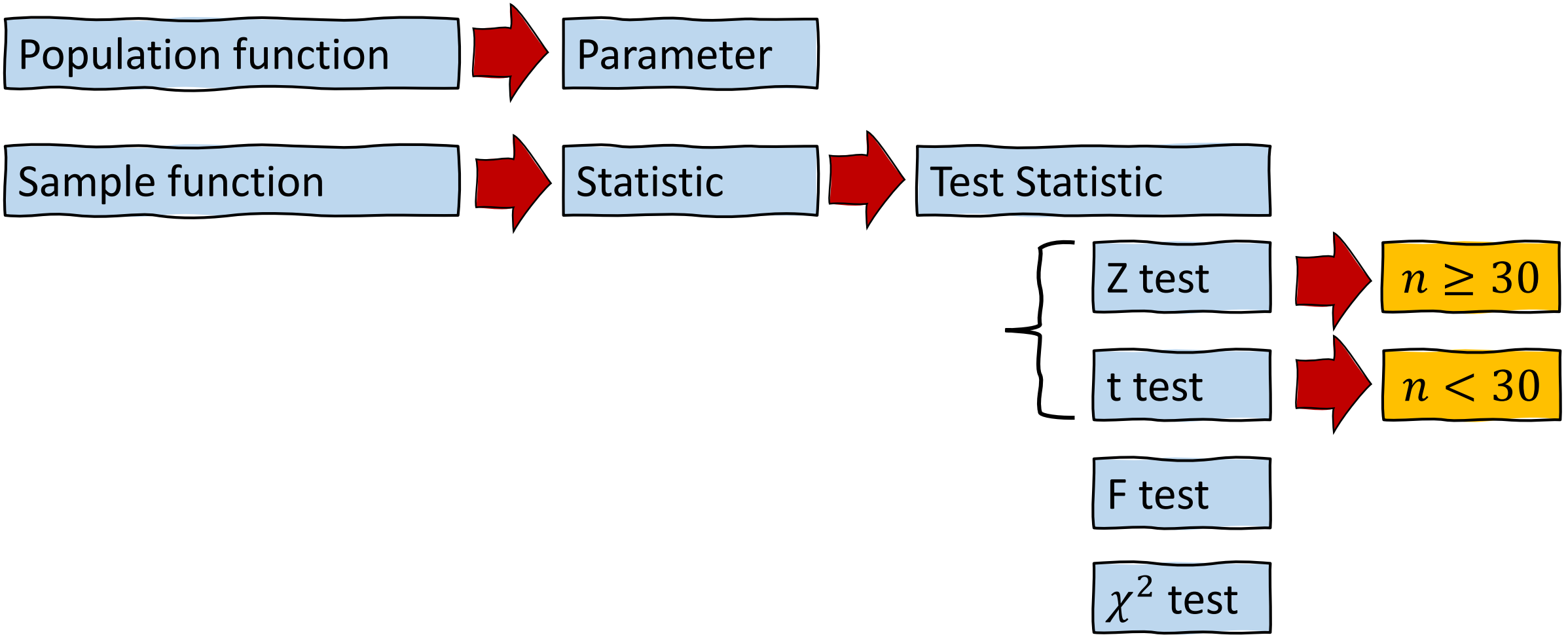
- A school reported that the average test score of students in 2023 was 85. We want to test if the average test score is 85.

$$H_0: \mu = 85$$

$$H_1: \mu \neq 85$$

Test Statistic

Verify on the basis of sample information



Example

- A sample of 120 users spends an average of 6.2 minutes on a website, with a standard deviation of 1.1 minutes. Test if the mean time is the average of 6.5 minutes.

$H_0: \mu = 6.5$

$H_1: \mu \neq 6.5$

$z\ test$

Example

- A company claims the average lifespan of its mobile batteries is 500 hours. A sample of 25 batteries shows an average lifespan of 495 hours with a standard deviation of 10 hours. Test whether the average lifespan is less than 500 hours

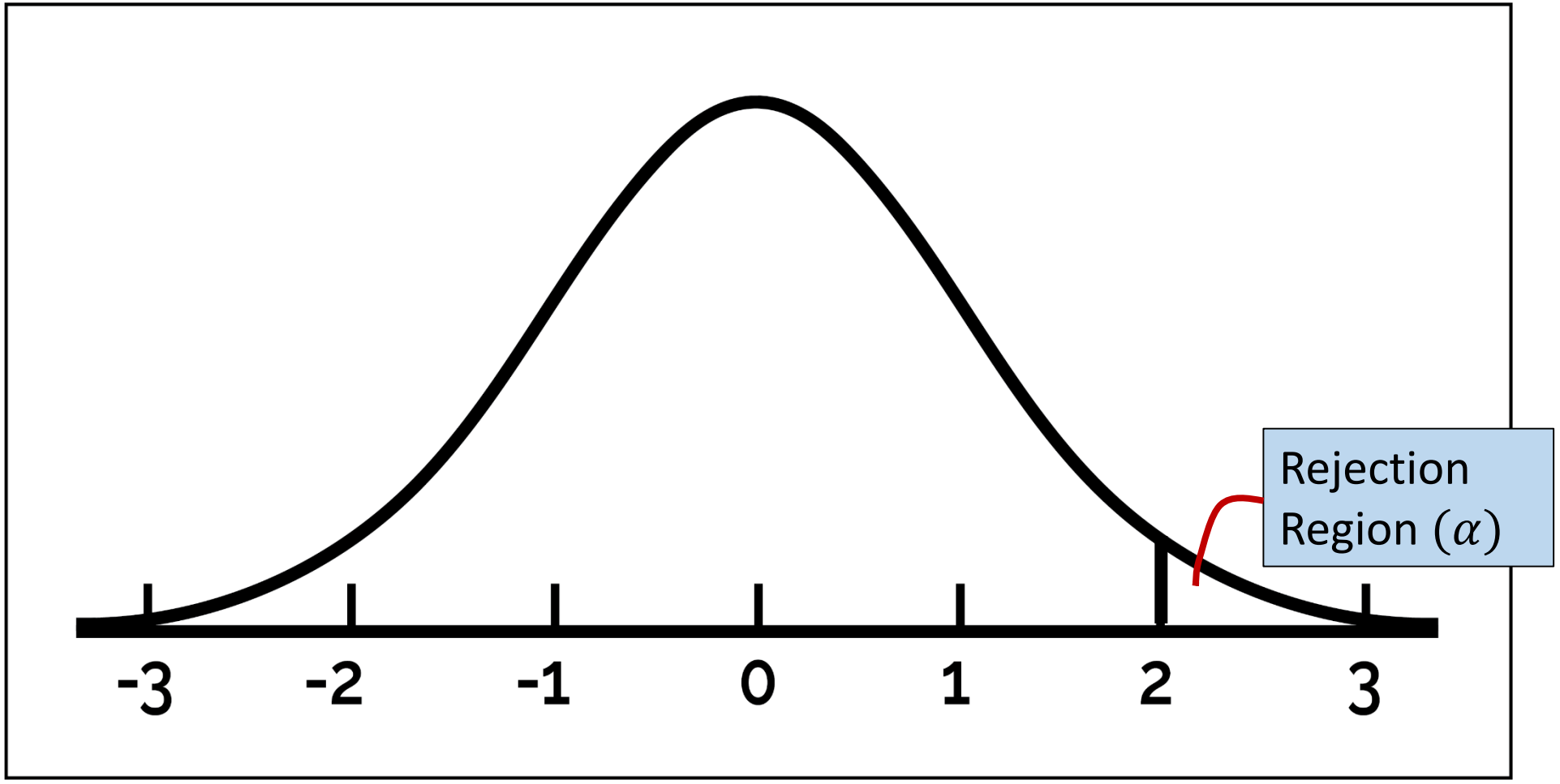
$H_0: \mu \geq 500$

$H_1: \mu < 500$

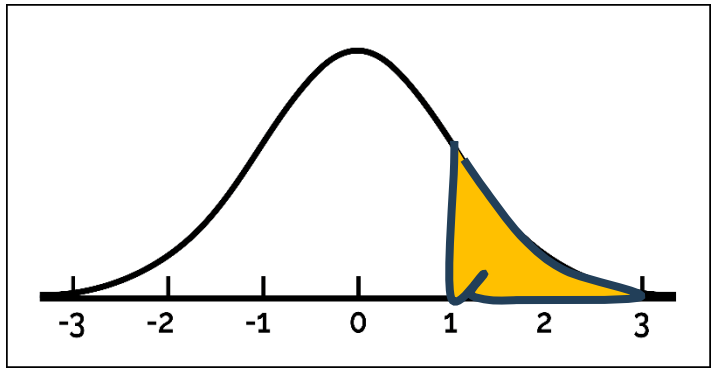
t *test*

Rejection Region

The region where we may reject our hypothesis.



Type of tailed



Right tailed test (one tailed)

$H_1: \mu > \mu_0$

- There are two types:

a. One tailed test

Entire rejection region lies one side

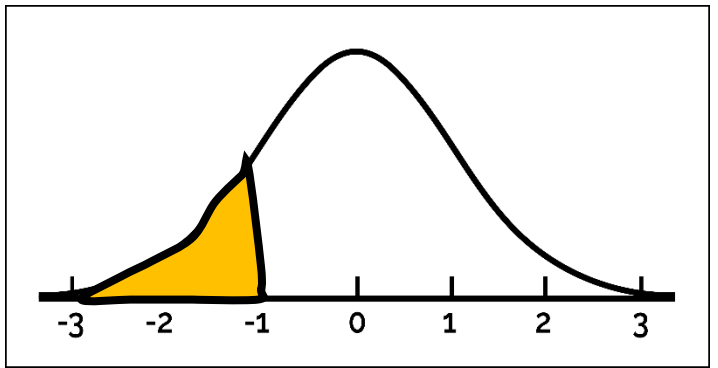
$H_1: \mu \neq \mu_0$

Entire rejection region lies two side

b. Two tailed test

Left tailed test (one tailed)

$H_1: \mu < \mu_0$



Example

- A sample of 120 users spends an average of 6.2 minutes on a website, with a standard deviation of 1.1 minutes. Test if the mean time is the claimed average of 6.5 minutes.

$H_0: \mu = 6.5$

$H_1: \mu \neq 6.5$

Two Tailed Test

Example

- A company claims the average lifespan of its mobile batteries is 500 hours. A sample of 25 batteries shows an average lifespan of 495 hours with a standard deviation of 10 hours. Test whether the average lifespan is less than 500 hours

$H_0: \mu \geq 500$

$H_1: \mu < 500$

Left Tailed Test

Steps (Z Test)

$$t = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$$

$$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

1. Identify the hypotheses
2. Choose the value of α; level of significance
3. Appropriate test statistic (Z *or* t) and determine the value.
4. Compare this value with critical value obtained by using α
5. Decision

Decision: We may reject H_0 if,

a) For left tailed: $Z_{cal} \leq -(Z_\alpha)$
b) For right tailed: $Z_{cal} \geq +(Z_\alpha)$

c) For two tailed: $|Z_{cal}| \geq Z_{\frac{\alpha}{2}}$

Critical value: When $\alpha = 0.05$

a) For one tailed: $Z_\alpha = 1.645$
b) For two tailed: $Z_{\frac{\alpha}{2}} = 1.96$

Critical value: When $\alpha = 0.01$

a) For one tailed: $Z_\alpha = 2.33$
b) For two tailed: $Z_{\frac{\alpha}{2}} = 2.56$

Example

$$H_0: \mu = 6.5$$

$$H_1: \mu \neq 6.5$$

$$\alpha = 0.05$$

- A sample of 120 users spends an average of 6.2 minutes on a website, with a standard deviation of 1.1 minutes. Test if the mean time is the average of 6.5 minutes.

$$Z_{cal} = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

$$Z_{cal} = \frac{6.2 - 6.5}{\frac{1.1}{\sqrt{120}}} = -3$$

Example

$H_0: \mu = 6.5$

$H_1: \mu \neq 6.5$

$\alpha = 0.05$

$$Z_{cal} = \frac{6.2 - 6.5}{\frac{1.1}{\sqrt{120}}} = -3$$

- A sample of 120 users spends an average of 6.2 minutes on a website, with a standard deviation of 1.1 minutes. Test if the mean time is the average of 6.5 minutes.

$Z_{tab} = 1.96$

$|Z_{cal}| > Z_{tab}$

$Reject\ H_0$

Decision: We may reject H_0 if,

a) For left tailed: $Z_{cal} \leq -Z_{\alpha}$
b) For right tailed: $Z_{cal} \geq +Z_{\alpha}$
c) For two tailed: $|Z_{cal}| \geq Z_{\frac{\alpha}{2}}$

Critical value: When $\alpha = 0.05$

a) For one tailed: $Z_{\alpha} = 1.645$
b) For two tailed: $Z_{\frac{\alpha}{2}} = 1.96$

Critical value: When $\alpha = 0.01$

a) For one tailed: $Z_{\alpha} = 2.33$
b) For two tailed: $Z_{\frac{\alpha}{2}} = 2.56$

Example

- The average monthly rent for apartments in a city is 1200 dollars. A sample of 81 apartments shows a mean rent of 1185, with a standard deviation 100 dollars. Test at the 5% significance level whether the sample mean differs from the population mean.

Example

- A study suggests the mean daily screen time for adults is 6 hours. A random sample of 66 adults has a mean daily screen time of 5.8 hours, with a standard deviation of 1.2 hours. Test at the 5% significance level whether screen time is less than 6 hours.

Thank You !